OAVS • TGT COMPUTER TEACHER EXAMINATION

SQL Commands

MCQ Question Bank

Database & Table Commands • DML Commands • Aggregate Functions
GROUP BY • ORDER BY • HAVING

108	10	100%
Total MCQs	Sections	Syllabus Match

Every question includes four options, the correct answer strip, and a short explanation.

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SECTION 1 OF 10

Database & Table Commands

Q1 **COMMAND IDENTIFICATION | DDL**

Which command creates a new database?

A. CREATE DATABASE

B. MAKE DATABASE

C. ADD DATABASE

D. NEW DATABASE

Ans: A

Explanation: CREATE DATABASE is the standard SQL command used to create a new database.

Q2 **COMMAND PURPOSE | DDL**

Which command permanently deletes an entire database along with all its tables?

A. DELETE DATABASE

B. DROP DATABASE

C. REMOVE DATABASE

D. TRUNCATE DATABASE

Ans: B

Explanation: DROP DATABASE removes the database along with all the tables and data stored in it.

Q3 **SYNTAX | DDL**

Which is the correct syntax to create a new table named Student?

A. NEW TABLE Student (...columns...);

B. CREATE Student TABLE (...columns...);

C. CREATE TABLE Student (...columns...);

D. TABLE CREATE Student (...columns...);

Ans: C

Explanation: The correct syntax begins with CREATE TABLE, followed by the table name and column definitions in brackets.

Q4 **SYNTAX | ALTER TABLE**

Which keyword is used with ALTER TABLE to add a new column to an existing table?

A. INSERT COLUMN

B. NEW COLUMN

C. CREATE COLUMN

D. ADD

Ans: D

Explanation: ALTER TABLE ... ADD ColumnName Datatype; is used to add a new column to an existing table.

Q5 **COMMAND IDENTIFICATION | DDL**

Which command is used to remove an entire table, including its structure and all data?

A. DROP TABLE

B. CLEAR TABLE

C. REMOVE TABLE

D. DELETE TABLE

Ans: A

Explanation: DROP TABLE deletes both the table structure and its stored data permanently.

Q6 **VALID/INVALID STATEMENT | ALTER TABLE**

Which of the following is a valid SQL statement to add a column 'Age' to table 'Student'?

A. INSERT COLUMN Age INTO Student;

B. ALTER TABLE Student ADD Age INT;

C. MODIFY TABLE Student ADD Age INT;

D. ALTER Student ADD COLUMN Age INT;

Ans: B

Explanation: The valid syntax is ALTER TABLE followed by the table name, ADD, and the new column with its datatype.

Q7 **COMMAND PURPOSE | DDL**

Which command is used to change the structure of an existing table, such as adding, modifying, or dropping a column?

A. MODIFY TABLE

B. CHANGE TABLE

C. ALTER TABLE

D. UPDATE TABLE

Ans: C

Explanation: ALTER TABLE is the DDL command used to change the structure of an existing table.

Q8 **SYNTAX | ALTER TABLE**

Which statement correctly deletes only the "Marks" column from table "Result"?

A. DELETE COLUMN Marks FROM Result;

B. REMOVE Marks FROM Result;

C. DROP COLUMN Marks;

D. ALTER TABLE Result DROP COLUMN Marks;

Ans: D

Explanation: ALTER TABLE ... DROP COLUMN is used to remove a specific column from an existing table.

Q9 **COMMAND PURPOSE | ALTER TABLE**

Which command is used to change the data type of an existing column in a table?

A. ALTER TABLE ... MODIFY

B. SET COLUMN TYPE

C. CHANGE TABLE ... TYPE

D. UPDATE TABLE ... CHANGE

Ans: A

Explanation: ALTER TABLE with the MODIFY clause is used to change the data type or size of an existing column.

Q10 **STATEMENT-BASED | DDL**

"CREATE DATABASE and CREATE TABLE are both DDL (Data Definition Language) commands used to define database structure." Is this statement correct?

A. False, CREATE TABLE is DML

B. True, both define structure

C. False, both are DML

D. True, but only CREATE DATABASE is DDL

Ans: B

Explanation: Both commands define the structure of database objects, so both belong to the DDL category.

Q11 **TRICKY | DDL**

Which statement correctly differentiates DROP DATABASE from DROP TABLE?

A. DROP DATABASE removes the entire database (all tables); DROP TABLE removes only one specified table

B. DROP DATABASE keeps the data but removes structure; DROP TABLE removes everything permanently

C. Both commands do exactly the same thing

D. DROP DATABASE removes only one table; DROP TABLE removes the entire database

Ans: A

Explanation: DROP DATABASE deletes the whole database along with every table in it, while DROP TABLE deletes just the one table specified.

Q12 **SYNTAX | CREATE TABLE**

Which statement correctly creates a table named Student with RollNo as an integer and Name as a 30-character text field?

A. CREATE TABLE Student (RollNo INT, Name VARCHAR(30));

B. CREATE TABLE Student (RollNo INT Name VARCHAR(30));

C. CREATE Student TABLE (RollNo INT, Name VARCHAR(30));

D. TABLE CREATE Student (RollNo INT, Name VARCHAR(30));

Ans: A

Explanation: Column definitions are placed in brackets after the table name, separated by commas, each with its own datatype.

SECTION 2 OF 10

DML Commands

Q13 **COMMAND IDENTIFICATION | DML**

Which of the following is NOT a DML command as per this syllabus?

A. DELETE

B. INSERT

C. CREATE

D. UPDATE

Ans: C

Explanation: CREATE defines database structure (DDL), while INSERT, UPDATE and DELETE manipulate data (DML).

Q14 **TERMINOLOGY | DML**

DML stands for:

A. Data Management Language

B. Data Modify Language

C. Database Modeling Language

D. Data Manipulation Language

Ans: D

Explanation: DML stands for Data Manipulation Language, used to manipulate data stored in tables.

Q15 **COMMAND PURPOSE | DML**

Which command is used to modify existing records in a table?

A. UPDATE

B. INSERT

C. SELECT

D. CREATE

Ans: A

Explanation: UPDATE is used to change the values of existing records in a table.

Q16 **COMMAND PURPOSE | DML**

Which command retrieves data from one or more tables?

A. FETCH

B. SELECT

C. SHOW

D. GET

Ans: B

Explanation: SELECT is the SQL command used to retrieve/query data from tables.

Q17 **MATCH THE FOLLOWING | DML**

Match the DML commands with their function:
Column A: 1. INSERT 2. UPDATE 3. DELETE 4. SELECT
Column B: (i) Retrieve records (ii) Add new records (iii) Modify records (iv) Remove records

A. 1-iii, 2-i, 3-ii, 4-iv

B. 1-ii, 2-iv, 3-iii, 4-i

C. 1-ii, 2-iii, 3-iv, 4-i

D. 1-i, 2-ii, 3-iii, 4-iv

Ans: C

Explanation: INSERT adds records, UPDATE modifies them, DELETE removes them, and SELECT retrieves them.

Q18 **ASSERTION-REASON | DML**

Assertion (A): DELETE removes rows from a table without deleting the table structure.
Reason (R): DELETE is a DML command that operates on table data, not on table definition.

A. A is true, R is false

B. Both A and R are true, but R does not explain A

C. A is false, R is true

D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: DELETE only removes data (rows) using DML, while the table structure defined by DDL remains unaffected.

SECTION 3 OF 10

INSERT, UPDATE, DELETE & SELECT

Q19 **SYNTAX | INSERT**

Which is the correct syntax to insert values into all columns of table Student(RollNo, Name, Marks)?

A. INSERT INTO Student VALUES (1,'Anu',90);

B. ADD INTO Student VALUES (1,'Anu',90);

C. INSERT Student VALUES (1,'Anu',90);

D. INSERT INTO VALUES Student (1,'Anu',90);

Ans: A

Explanation: The correct syntax is INSERT INTO TableName VALUES(...) when inserting values for all columns in order.

Q20 **SYNTAX | UPDATE**

Which clause is essential with UPDATE to change only specific record(s) instead of the entire table?

A. HAVING

B. WHERE

C. ORDER BY

D. GROUP BY

Ans: B

Explanation: The WHERE clause specifies the condition to identify which record(s) should be updated.

Q21 **OUTPUT-BASED | UPDATE**

What happens if the UPDATE command is executed without a WHERE clause?

A. A syntax error occurs

B. Only the first row is updated

C. All rows in the table get updated

D. The command is ignored

Ans: C

Explanation: Without a WHERE clause, UPDATE applies the change to every row in the table.

Q22 **COMMAND IDENTIFICATION | DELETE**

Which command is used to remove specific row(s) from a table based on a condition?

A. TRUNCATE

B. REMOVE

C. DROP

D. DELETE

Ans: D

Explanation: DELETE removes rows that satisfy a specified condition while keeping the table structure intact.

Q23 **VALID/INVALID STATEMENT | SELECT**

Which is the correct SELECT statement to display Name and Marks of all students?

A. SELECT Name, Marks FROM Student;

B. SELECT * Name, Marks Student;

C. GET Name, Marks FROM Student;

D. SELECT Name Marks FROM Student;

Ans: A

Explanation: Column names must be comma-separated after SELECT, followed by FROM and the table name.

Q24 **QUERY-BASED | UPDATE**

In table Employee(EmpID, Name, Salary), which statement increases the Salary of employee EmpID=5 by 1000?

A. UPDATE SET Salary=Salary+1000 FROM Employee WHERE EmpID=5;

B. UPDATE Employee SET Salary=Salary+1000 WHERE EmpID=5;

C. ALTER Employee SET Salary=Salary+1000;

D. UPDATE Employee Salary=Salary+1000 WHERE EmpID=5;

Ans: B

Explanation: UPDATE requires the SET keyword before assigning new values and a WHERE clause to target the correct row.

Q25 **SYNTAX | SELECT**

Which symbol is used in a SELECT statement to fetch all columns of a table?

A. %

B. #

C. *

D. ALL

Ans: C

Explanation: The asterisk (*) symbol represents "all columns" in a SELECT statement.

Q26 **ERROR IDENTIFICATION | DELETE**

What does the statement "DELETE FROM Student WHERE RollNo=5;" do?

A. Causes an error since WHERE is not allowed with DELETE

B. Deletes the entire Student table

C. Deletes only the RollNo column

D. Deletes only the row where RollNo is 5

Ans: D

Explanation: The WHERE clause restricts DELETE to only the row(s) matching the given condition.

Q27 **COMMAND PURPOSE | INSERT**

Which command would you use to add a new student's record without affecting existing records?

A. INSERT

B. ALTER

C. CREATE

D. UPDATE

Ans: A

Explanation: INSERT adds new rows to a table without changing the existing data.

Q28 **SCENARIO-BASED | UPDATE**

In table Library(BookID, Title, Status), which statement correctly changes the Status of BookID=101 to 'Issued'?

A. INSERT INTO Library SET Status='Issued' WHERE BookID=101;

B. UPDATE Library SET Status='Issued' WHERE BookID=101;

C. ALTER Library SET Status='Issued' WHERE BookID=101;

D. UPDATE Library WHERE BookID=101 SET Status='Issued';

Ans: B

Explanation: The correct UPDATE syntax places SET before WHERE, assigning the new value before specifying the condition.

Q29 **SYNTAX | INSERT**

Which statement correctly inserts a value into only the Name and Marks columns of table Student(RollNo, Name, Marks), leaving RollNo to be filled later?

A. INSERT INTO Student VALUES ('Anu', 90);

B. INSERT INTO Student (Name, Marks) VALUES ('Anu', 90);

C. INSERT Student (Name, Marks) VALUES ('Anu', 90);

D. INSERT INTO Student SET (Name, Marks) = ('Anu', 90);

Ans: B

Explanation: Specifying the column list in parentheses after the table name lets you insert values for only selected columns.

Q30 **SYNTAX | UPDATE**

Which statement correctly updates both the Name and Marks of the student with RollNo=3 in one statement?

A. UPDATE Student SET Name='Rohit' AND Marks=85 WHERE RollNo=3;

B. UPDATE Student SET Name='Rohit', SET Marks=85 WHERE RollNo=3;

C. UPDATE Student SET Name='Rohit', Marks=85 WHERE RollNo=3;

D. UPDATE Student SET Name='Rohit' SET Marks=85 WHERE RollNo=3;

Ans: C

Explanation: Multiple columns can be updated in a single UPDATE statement by separating the column=value pairs with commas after SET.

SECTION 4 OF 10

Aggregate Functions

Q31 **COMMAND IDENTIFICATION | AGGREGATE**

Which function is used to calculate the total sum of values in a numeric column?

A. COUNT()

B. AVG()

C. SUM()

D. MAX()

Ans: C

Explanation: SUM() adds up all the numeric values in the specified column.

Q32 **COMMAND IDENTIFICATION | AGGREGATE**

Which aggregate function returns the highest value in a column?

A. HIGH()

B. MIN()

C. TOP()

D. MAX()

Ans: D

Explanation: MAX() returns the largest value present in the specified column.

Q33 **COMMAND IDENTIFICATION | AGGREGATE**

Which aggregate function returns the lowest value in a column?

A. MIN()

B. LOW()

C. SMALL()

D. BOTTOM()

Ans: A

Explanation: MIN() returns the smallest value present in the specified column.

Q34 **COMMAND IDENTIFICATION | AGGREGATE**

Which function calculates the average (mean) value of a numeric column?

A. SUM()/COUNT()

B. AVG()

C. AVERAGE()

D. MEAN()

Ans: B

Explanation: AVG() computes the mean of all values in the specified numeric column.

Q35 **QUERY-BASED | AGGREGATE**

Which SQL function, used as COUNT(*), returns the total number of rows in a result set, including rows with NULL values?

A. COUNT(column)

B. SUM()

C. COUNT(*)

D. AVG()

Ans: C

Explanation: COUNT(*) counts all rows regardless of NULL values, unlike COUNT(column) which ignores NULLs in that column.

Q36 **QUERY-BASED | AGGREGATE**

In table Marks(StudentID, Subject, Score), which query finds the highest score obtained by any student?

A. SELECT TOP(Score) FROM Marks;

B. SELECT MAX(Score) FROM Marks GROUP BY Score;

C. SELECT HIGH(Score) FROM Marks;

D. SELECT MAX(Score) FROM Marks;

Ans: D

Explanation: MAX(Score) directly returns the single highest value present in the Score column.

Q37 **VALID/INVALID STATEMENT | AGGREGATE**

Which of the following is NOT a valid aggregate function under this syllabus?

A. TOTAL()

B. COUNT()

C. SUM()

D. AVG()

Ans: A

Explanation: TOTAL() is not a standard SQL aggregate function; SUM(), COUNT() and AVG() are the valid ones listed.

Q38 **OUTPUT-BASED | AGGREGATE**

Query: "SELECT COUNT(Marks) FROM Result WHERE Marks IS NOT NULL;" — what does this count?

A. Total sum of Marks

B. Number of rows where the Marks value is present (not NULL)

C. Number of rows with NULL Marks

D. Average of Marks

Ans: B

Explanation: COUNT(Marks) counts only the rows where Marks has a non-NULL value.

Q39 **QUERY-BASED | AGGREGATE**

Which query finds the total number of employees in table Employee(EmpID, Name, Dept)?

A. SELECT MAX(EmpID) FROM Employee;

B. SELECT AVG(EmpID) FROM Employee;

C. SELECT COUNT(*) FROM Employee;

D. SELECT SUM(EmpID) FROM Employee;

Ans: C

Explanation: COUNT(*) counts the total number of rows, which equals the total number of employees.

Q40 **ASSERTION-REASON | AGGREGATE**

Assertion (A): AVG() ignores NULL values while calculating the average of a column. Reason (R): Aggregate functions in SQL generally exclude NULL values from their calculation, except COUNT(*).

A. Both A and R are true, but R does not explain A

B. A is true, R is false

C. A is false, R is true

D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: Most aggregate functions skip NULL values during calculation, which correctly explains why AVG() excludes them.

Q41 **COMMAND PURPOSE | AGGREGATE**

If "SELECT AVG(Marks) FROM Student;" is run without a GROUP BY clause, what does it return?

A. A separate average for each row

B. An error, since AVG() requires GROUP BY

C. NULL, because GROUP BY is mandatory

D. A single average value calculated over all rows in the table

Ans: D

Explanation: Without GROUP BY, an aggregate function treats the entire table as one group and returns a single summary value.

Q42 **TRICKY | AGGREGATE**

If the Marks column has one NULL value among 5 rows, what is the difference between COUNT(*) and COUNT(Marks)?

A. COUNT(*) returns 4, while COUNT(Marks) returns 5

B. COUNT(*) returns 5, while COUNT(Marks) returns 4

C. Both return 4

D. Both return 5

Ans: B

Explanation: COUNT(*) counts every row including NULLs, while COUNT(column) counts only the rows where that column is not NULL.

SECTION 5 OF 10

GROUP BY

Q43 **COMMAND IDENTIFICATION | GROUP BY**

Which clause is used to arrange identical data into groups based on one or more columns?

A. GROUP BY

B. HAVING

C. ORDER BY

D. JOIN BY

Ans: A

Explanation: GROUP BY groups rows that share the same values in specified columns.

Q44 **QUERY-BASED | GROUP BY**

In table Sales(Region, Amount), which query gives the total sales amount for each region?

A. SELECT SUM(Amount) FROM Sales GROUP Region;

B. SELECT Region, SUM(Amount) FROM Sales GROUP BY Region;

C. SELECT Region, SUM(Amount) FROM Sales;

D. SELECT Region, SUM(Amount) FROM Sales ORDER BY Region;

Ans: B

Explanation: GROUP BY Region is required to calculate a separate SUM(Amount) for each region.

Q45 **COMMAND PURPOSE | GROUP BY**

GROUP BY is usually used together with which type of SQL functions?

A. Date functions

B. Mathematical functions only

C. Aggregate functions

D. String functions

Ans: C

Explanation: GROUP BY is commonly paired with aggregate functions like SUM(), COUNT(), AVG() to summarize each group.

Q46 **ERROR IDENTIFICATION | GROUP BY**

In the query "SELECT Dept, COUNT(*) FROM Employee;", intended to show employee count per department, what is missing?

A. Nothing is wrong

B. COUNT(*) is invalid

C. SELECT cannot use two columns

D. The GROUP BY Dept clause

Ans: D

Explanation: Without GROUP BY Dept, the query cannot correctly summarize COUNT(*) separately for each department.

Q47 **QUERY-BASED | GROUP BY**

In table Student(Class, Name, Marks), which query lists each Class along with the number of students in it?

A. SELECT Class, COUNT(*) FROM Student GROUP BY Class;

B. SELECT Class, COUNT(*) FROM Student;

C. SELECT COUNT(Class) FROM Student;

D. SELECT Class, COUNT(*) FROM Student ORDER BY Class;

Ans: A

Explanation: GROUP BY Class is needed so COUNT(*) is calculated separately for each class.

Q48 **COMMAND PURPOSE | GROUP BY**

When GROUP BY is used, which column(s) can safely appear in SELECT without an aggregate function?

A. None, all columns need aggregate functions

B. Only the column(s) listed in the GROUP BY clause

C. Any column of the table

D. Only numeric columns

Ans: B

Explanation: Non-aggregated columns in SELECT must match the columns used in GROUP BY to give meaningful results.

Q49 **OUTPUT-BASED | GROUP BY**

Query: "SELECT Dept, AVG(Salary) FROM Employee GROUP BY Dept;" — what does this return?

A. Total salary for each department

B. Highest salary in each department

C. Average salary for each department

D. Average salary of the whole company

Ans: C

Explanation: GROUP BY Dept causes AVG(Salary) to be calculated separately for every department.

Q50 **STATEMENT-BASED | GROUP BY**

"GROUP BY groups rows that have the same values in specified columns, and is often used with aggregate functions to summarize each group." Is this statement correct?

A. False, GROUP BY cannot be used with aggregate functions

B. False, GROUP BY works only on numeric columns

C. False, GROUP BY only sorts data

D. True

Ans: D

Explanation: This is a correct description of the purpose and typical use of the GROUP BY clause.

Q51 **QUERY-BASED | GROUP BY**

In table Sales(Region, Product, Amount), which query gives total Amount for each Region-Product combination?

A. SELECT Region, Product, SUM(Amount) FROM Sales GROUP BY Region, Product;

B. SELECT Region, Product, SUM(Amount) FROM Sales GROUP BY Region;

C. SELECT SUM(Amount) FROM Sales GROUP BY Region AND Product;

D. SELECT Region, Product, SUM(Amount) FROM Sales GROUP BY Amount;

Ans: A

Explanation: Listing both columns after GROUP BY, separated by a comma, groups rows by every unique Region-Product combination.

SECTION 6 OF 10

ORDER BY

Q52 **COMMAND IDENTIFICATION | ORDER BY**

Which clause is used to sort the result set of a query?

A. ORDER BY

B. SORT BY

C. GROUP BY

D. HAVING

Ans: A

Explanation: ORDER BY arranges the output rows of a query in ascending or descending order.

Q53 **COMMAND PURPOSE | ORDER BY**

By default, ORDER BY sorts records in which order?

A. No specific order

B. Ascending

C. Descending

D. Random

Ans: B

Explanation: If no keyword is specified, ORDER BY sorts the result in ascending order by default.

Q54 **SYNTAX | ORDER BY**

Which keyword is used with ORDER BY to sort records in descending order?

A. LOW

B. REV

C. DESC

D. DOWN

Ans: C

Explanation: DESC is added after the column name in ORDER BY to sort results from highest to lowest.

Q55 **QUERY-BASED | ORDER BY**

In table Student(Name, Marks), which query displays students sorted by Marks from highest to lowest?

A. SELECT Name, Marks FROM Student ORDER BY Marks;

B. SELECT Name, Marks FROM Student SORT BY Marks DESC;

C. SELECT Name, Marks FROM Student GROUP BY Marks DESC;

D. SELECT Name, Marks FROM Student ORDER BY Marks DESC;

Ans: D

Explanation: ORDER BY Marks DESC arranges the rows from the highest to the lowest Marks value.

Q56 **COMMAND PURPOSE | ORDER BY**

Can ORDER BY be used to sort by more than one column at the same time?

A. Yes, multiple columns can be specified, separated by commas

B. No, only a single column is allowed

C. Only when using SELECT *

D. Only when used with GROUP BY

Ans: A

Explanation: ORDER BY accepts a comma-separated list of columns, sorting first by the first column, then the next.

Q57 **SYNTAX | ORDER BY**

In the query "SELECT Name FROM Student ORDER BY Name ASC;", what does ASC represent?

A. Automatic sort

B. Ascending order

C. An aggregate sort condition

D. A special column

Ans: B

Explanation: ASC explicitly specifies that the results should be sorted in ascending order.

Q58 **COMMAND PURPOSE | ORDER BY**

Among GROUP BY, HAVING and ORDER BY, which clause is generally written last in a SELECT query?

A. All appear in the same position

B. HAVING

C. ORDER BY

D. GROUP BY

Ans: C

Explanation: ORDER BY is applied last, after grouping and filtering of groups is complete, to sort the final output.

Q59 **SCENARIO-BASED | ORDER BY**

A teacher wants the marks list arranged from lowest to highest. Which query is correct?

A. SELECT * FROM Marks ORDER BY Score DESC;

B. SELECT * FROM Marks GROUP BY Score ASC;

C. SELECT * FROM Marks ORDER BY Score HAVING ASC;

D. SELECT * FROM Marks ORDER BY Score;

Ans: D

Explanation: Since ascending is the default order, ORDER BY Score alone arranges data from lowest to highest.

Q60 **SYNTAX | ORDER BY**

In the query "SELECT Name, Marks FROM Student ORDER BY 2 DESC;", what does the number 2 refer to?

A. The second table in the FROM clause

B. The second column in the SELECT list, i.e. Marks

C. The second row of the table

D. A fixed numeric sort value

Ans: B

Explanation: In ORDER BY, a number refers to the position of the column as listed in the SELECT clause, so 2 means Marks.

SECTION 7 OF 10

HAVING

Q61 **COMMAND IDENTIFICATION | HAVING**

Which clause filters groups after aggregation, unlike WHERE which filters individual rows before grouping?

A. HAVING

B. ORDER BY

C. WHERE

D. GROUP BY

Ans: A

Explanation: HAVING is applied after GROUP BY to filter the aggregated groups based on a condition.

Q62 **COMMAND PURPOSE | HAVING**

The HAVING clause must generally be used along with which clause?

A. It can only be used alone

B. GROUP BY

C. SELECT DISTINCT

D. ORDER BY

Ans: B

Explanation: HAVING filters the groups created by GROUP BY, so it is typically used together with it.

Q63 **OUTPUT-BASED | HAVING**

Query: "SELECT Dept, COUNT(*) FROM Employee GROUP BY Dept HAVING COUNT(*) > 5;" — what does this return?

A. Departments with less than 5 employees

B. All departments regardless of employee count

C. Departments with more than 5 employees

D. Employees with more than 5 records

Ans: C

Explanation: HAVING COUNT(*) > 5 keeps only those grouped departments whose employee count exceeds 5.

Q64 **VALID/INVALID STATEMENT | HAVING**

Which of the following correctly describes a valid use of HAVING?

A. HAVING can filter individual rows before any grouping takes place

B. HAVING is used only to sort grouped results

C. HAVING replaces the SELECT clause

D. HAVING filters groups based on a condition applied to an aggregate function

Ans: D

Explanation: HAVING is specifically designed to apply conditions on aggregated group values, unlike WHERE.

Q65 **ERROR IDENTIFICATION | HAVING**

In the query "SELECT Region, SUM(Amount) FROM Sales HAVING SUM(Amount) > 10000;", what clause is missing?

A. GROUP BY Region

B. ORDER BY clause

C. Nothing is missing

D. WHERE clause

Ans: A

Explanation: HAVING needs groups to filter, so GROUP BY Region must be included before HAVING can work correctly.

Q66 **ASSERTION-REASON | HAVING**

Assertion (A): HAVING can filter records using aggregate functions, but WHERE cannot.
Reason (R): WHERE is applied before grouping/aggregation, so aggregate results are not yet available to it.

A. A is true, R is false

B. Both A and R are true, and R correctly explains A

C. A is false, R is true

D. Both A and R are true, but R does not explain A

Ans: B

Explanation: Since WHERE executes before aggregation, it cannot filter on aggregate results, which is exactly why HAVING is needed.

Q67 **OUTPUT-BASED | HAVING**

Query: "SELECT Class, AVG(Marks) FROM Student GROUP BY Class HAVING AVG(Marks) >= 75;" — which classes are shown in the result?

A. All classes irrespective of marks

B. Only the class with the single highest marks

C. Classes with average marks of 75 or above

D. Classes with average marks less than 75

Ans: C

Explanation: HAVING AVG(Marks) >= 75 retains only those class-groups whose average marks meet or exceed 75.

Q68 **SYNTAX | HAVING**

Which is the correct clause order in a SELECT query using GROUP BY, HAVING and ORDER BY together?

A. SELECT ... FROM ... HAVING ... ORDER BY ... GROUP BY

B. SELECT ... FROM ... HAVING ... GROUP BY ... ORDER BY

C. SELECT ... FROM ... ORDER BY ... GROUP BY ... HAVING

D. SELECT ... FROM ... GROUP BY ... HAVING ... ORDER BY

Ans: D

Explanation: SQL executes grouping first, then filters groups with HAVING, and finally sorts the result with ORDER BY.

Q69 **QUERY-BASED | HAVING**

Which query correctly shows departments with more than 3 employees AND an average salary above 40000?

A. SELECT Dept, COUNT(*), AVG(Salary) FROM Employee GROUP BY Dept HAVING COUNT(*) > 3 OR AVG(Salary) > 40000 ONLY;

B. SELECT Dept, COUNT(*), AVG(Salary) FROM Employee HAVING COUNT(*) > 3 AND AVG(Salary) > 40000;

C. SELECT Dept, COUNT(*), AVG(Salary) FROM Employee GROUP BY Dept HAVING COUNT(*) > 3 AND AVG(Salary) > 40000;

D. SELECT Dept, COUNT(*), AVG(Salary) FROM Employee GROUP BY Dept WHERE COUNT(*) > 3 AND AVG(Salary) > 40000;

Ans: C

Explanation: Multiple conditions can be combined in HAVING using AND, and it must follow a GROUP BY clause to filter aggregated groups.

SECTION 8 OF 10

Query / Output-Based Questions

REFERENCE TABLE — STUDENT (USED FOR SECTION 8 QUERY/OUTPUT-BASED QUESTIONS)

RollNo	Name	Class	Marks
1	Aman	10	78
2	Riya	10	92
3	Sohan	9	65
4	Meena	9	88
5	Karan	10	55

Q70 **OUTPUT-BASED | TABLE QUERY**

Based on the Student table above, what is the output of "SELECT COUNT(*) FROM Student;"?

A. 5

B. 4

C. 6

D. 3

Ans: A

Explanation: The table contains 5 rows in total, so COUNT(*) returns 5.

Q71 **OUTPUT-BASED | AGGREGATE**

Based on the Student table above, what is the output of "SELECT MAX(Marks) FROM Student;"?

A. 78

B. 92

C. 88

D. 65

Ans: B

Explanation: The highest Marks value in the table is 92, scored by Riya.

Q72 **OUTPUT-BASED | AGGREGATE**

Based on the Student table above, what is the output of "SELECT MIN(Marks) FROM Student;"?

A. 88

B. 78

C. 55

D. 65

Ans: C

Explanation: The lowest Marks value in the table is 55, scored by Karan.

Q73 **OUTPUT-BASED | GROUP BY**

Based on the Student table above, what is the output of "SELECT Class, COUNT(*) FROM Student GROUP BY Class;"?

A. Class 9 → 3, Class 10 → 2

B. Class 9 → 1, Class 10 → 4

C. Class 9 → 2, Class 10 → 2

D. Class 9 → 2, Class 10 → 3

Ans: D

Explanation: Class 9 has 2 students (Sohan, Meena) and Class 10 has 3 students (Aman, Riya, Karan).

Q74 **OUTPUT-BASED | AGGREGATE**

Based on the Student table above, what is the output of "SELECT SUM(Marks) FROM Student WHERE Class=10;"?

A. 225

B. 378

C. 245

D. 153

Ans: A

Explanation: For Class 10, Marks are 78+92+55, which sum to 225.

Q75 **OUTPUT-BASED | ORDER BY**

Based on the Student table above, if results are sorted using "ORDER BY Marks;" (ascending), which student's name appears first?

A. Riya

B. Karan

C. Meena

D. Aman

Ans: B

Explanation: Karan has the lowest Marks (55), so he appears first in ascending order.

Q76 **OUTPUT-BASED | AGGREGATE**

Based on the Student table above, what is the output of "SELECT AVG(Marks) FROM Student WHERE Class=9;"?

A. 78

B. 74

C. 76.5

D. 80

Ans: C

Explanation: For Class 9, Marks are 65 and 88; their average is (65+88)/2 = 76.5.

Q77 **OUTPUT-BASED | HAVING**

Based on the Student table above, what is the output of "SELECT Class, AVG(Marks) FROM Student GROUP BY Class HAVING AVG(Marks) > 75;"?

A. Both classes are shown

B. Only Class 10 is shown

C. Neither class is shown

D. Only Class 9 is shown

Ans: D

Explanation: Class 9's average is 76.5 (>75), while Class 10's average is exactly 75, so only Class 9 satisfies HAVING.

Q78 **OUTPUT-BASED | UPDATE**

Based on the Student table above, if "UPDATE Student SET Marks=100 WHERE RollNo=5;" is executed, what will Karan's new Marks value be?

A. 100

B. NULL

C. 55

D. 0

Ans: A

Explanation: The UPDATE statement sets Marks to 100 specifically for the row where RollNo equals 5 (Karan).

Q79 **OUTPUT-BASED | DELETE**

Based on the Student table above, if "DELETE FROM Student WHERE Class=9;" is executed, how many rows remain?

A. 5

B. 3

C. 0

D. 2

Ans: B

Explanation: The two Class 9 rows (Sohan, Meena) are deleted, leaving the 3 Class 10 rows.

Q80 **OUTPUT-BASED | ORDER BY**

Based on the Student table above, what is the output of "SELECT Name FROM Student ORDER BY Marks DESC;" for the very first name listed?

A. Aman

B. Sohan

C. Meena

D. Riya

Ans: D

Explanation: Riya has the highest Marks (92), so she appears first when sorted in descending order.

Q81 **OUTPUT-BASED | TABLE QUERY**

Based on the Student table above, what is the output of "SELECT COUNT(*) FROM Student WHERE Marks > 70;"?

A. 3

B. 5

C. 2

D. 4

Ans: A

Explanation: Three students — Aman(78), Riya(92), Meena(88) — have Marks greater than 70.

SECTION 9 OF 10

Important & Tricky Questions

Q82 **TRICKY | DDL VS DML**

Which statement correctly differentiates DROP TABLE from DELETE (without WHERE)?

A. DROP TABLE removes only data; DELETE removes structure

B. Both remove only data and keep the structure

C. DROP TABLE removes structure and data; DELETE removes only data, and structure remains

D. There is no difference between them

Ans: C

Explanation: DROP TABLE is a DDL command that erases the table entirely, whereas DELETE is a DML command that only clears data.

Q83 **TRICKY | DDL**

Which command would completely remove a table so that even its structure no longer exists in the database?

A. DELETE FROM TableName;

B. TRUNCATE TableName;

C. ALTER TABLE TableName DROP;

D. DROP TABLE TableName;

Ans: D

Explanation: DROP TABLE eliminates both the data and the table's structure permanently.

Q84 **MATCH THE FOLLOWING | MIXED**

Match the Following:
Column A: 1. SUM() 2. GROUP BY 3. HAVING 4. ORDER BY
Column B: (i) Filters groups after aggregation (ii) Sorts the final result (iii) Adds values of a column (iv) Groups rows with the same values

A. 1-iii, 2-iv, 3-i, 4-ii

B. 1-i, 2-ii, 3-iii, 4-iv

C. 1-iv, 2-iii, 3-ii, 4-i

D. 1-ii, 2-i, 3-iv, 4-iii

Ans: A

Explanation: SUM() adds values, GROUP BY groups rows, HAVING filters groups, and ORDER BY sorts the final result.

Q85 **ASSERTION-REASON | MIXED**

Assertion (A): ORDER BY can be used without GROUP BY, but HAVING generally requires GROUP BY.
Reason (R): ORDER BY only sorts the final output, whereas HAVING filters aggregated group results.

A. Both A and R are true, but R does not explain A

B. Both A and R are true, and R correctly explains A

C. Both A and R are false

D. A is true, R is false

Ans: B

Explanation: Since HAVING depends on groups formed by GROUP BY, while ORDER BY just sorts output, the reason correctly justifies the assertion.

Q86 **TRICKY | CLAUSE ORDER**

In a query using GROUP BY, HAVING and ORDER BY together, which is the correct clause order?

A. SELECT, FROM, HAVING, GROUP BY, ORDER BY

B. SELECT, FROM, ORDER BY, GROUP BY, HAVING

C. SELECT, FROM, GROUP BY, HAVING, ORDER BY

D. SELECT, FROM, GROUP BY, ORDER BY, HAVING

Ans: C

Explanation: SQL processes grouping, then filters groups with HAVING, and finally sorts the output using ORDER BY.

Q87 **ERROR IDENTIFICATION | GROUP BY**

Why is "SELECT Name, Marks FROM Student GROUP BY Class;" considered problematic?

A. SELECT cannot use commas between column names

B. The Class column doesn't exist

C. GROUP BY cannot be used with SELECT

D. Name and Marks are not aggregated and are not part of the GROUP BY column, causing inconsistent results

Ans: D

Explanation: When GROUP BY is used, non-grouped, non-aggregated columns like Name and Marks produce ambiguous or inconsistent output.

Q88 **TRICKY | AGGREGATE**

If table Employee has 0 rows, what will "SELECT COUNT(*) FROM Employee;" return?

A. 0

B. Blank/no output

C. Error

D. NULL

Ans: A

Explanation: COUNT(*) always returns a numeric value, and it correctly returns 0 when there are no rows to count.

Q89 **TRICKY | AGGREGATE**

If table Employee has 0 rows, what will "SELECT SUM(Salary) FROM Employee;" return?

A. Error

B. NULL

C. Blank table

D. 0

Ans: B

Explanation: Unlike COUNT(*), SUM() returns NULL when there are no rows to sum, since there are no values to add.

Q90 **TRICKY | SELECT**

Which of the following commands does not strictly require a WHERE clause for normal use, but can still accept one to restrict its result?

A. CREATE DATABASE

B. DROP TABLE

C. SELECT

D. CREATE TABLE

Ans: C

Explanation: SELECT can be used with or without WHERE, while commands like CREATE and DROP don't use WHERE at all.

Q91 **ASSERTION-REASON | MIXED**

Assertion (A): CREATE TABLE and ALTER TABLE both fall under DDL, while INSERT and UPDATE fall under DML.
Reason (R): DDL commands define or modify the structure of database objects, while DML commands manipulate the data stored within them.

A. A is false, R is true

B. Both A and R are true, but R does not explain A

C. A is true, R is false

D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: This is a correct and complete explanation of how DDL and DML commands are categorized.

Q92 **ERROR IDENTIFICATION | TRICKY**

Why is the statement "SELECT Dept FROM Employee WHERE COUNT(*) > 5;" invalid?

A. Dept is not a valid column name

B. Aggregate functions like COUNT() cannot be used directly in a WHERE clause

C. COUNT() can only be used with INSERT

D. WHERE cannot be used with SELECT

Ans: B

Explanation: WHERE filters rows before aggregation happens, so aggregate functions must be used in HAVING instead, after GROUP BY.

Q93 **TRICKY | GROUP BY**

What is the output of "SELECT Class FROM Student GROUP BY Class;" on the Student table, even though no aggregate function is used?

A. It causes a syntax error since no aggregate function is used

B. It returns only the first row's Class value

C. It lists each distinct Class value once (9 and 10)

D. It lists all 5 rows unchanged

Ans: C

Explanation: GROUP BY can be used alone to collapse rows into unique groups, effectively listing each distinct value once, even without an aggregate function.

SECTION 10 OF 10

Mixed OAVS TGT Practice Set

Q94 **COMMAND PURPOSE | DDL**

Which command must be used first, before data can be inserted into a newly designed table?

A. CREATE TABLE

B. UPDATE

C. INSERT

D. SELECT

Ans: A

Explanation: A table's structure must be created first using CREATE TABLE before any data can be inserted into it.

Q95 **COMMAND IDENTIFICATION | MIXED**

Which of the following pairs correctly matches a command with its category?

A. CREATE TABLE - DML

B. DELETE - DML

C. SELECT - DDL

D. DROP TABLE - DML

Ans: B

Explanation: DELETE manipulates data and belongs to DML, whereas CREATE TABLE and DROP TABLE belong to DDL.

Q96 **QUERY-BASED | AGGREGATE**

Query: "SELECT COUNT(*) FROM Book WHERE Price > 500;" — what does this determine?

A. The maximum price among books

B. The average price of books

C. The number of books priced above 500

D. The total price of all books

Ans: C

Explanation: COUNT(*) with the WHERE condition counts only the rows where the Price exceeds 500.

Q97 **SCENARIO-BASED | MIXED**

A school database needs a report of subject-wise average marks, but only for subjects where more than 20 students appeared. Which clauses should be combined?

A. ORDER BY and HAVING only

B. WHERE and ORDER BY only

C. DROP and ALTER

D. GROUP BY and HAVING

Ans: D

Explanation: GROUP BY groups records by subject, and HAVING filters those groups based on the aggregated student count.

Q98 **COMMAND PURPOSE | DDL**

Which SQL command permanently removes the "Marks" table and all its data from the database?

A. DROP TABLE Marks;

B. TRUNCATE Marks;

C. ALTER TABLE Marks DROP;

D. DELETE Marks;

Ans: A

Explanation: DROP TABLE Marks; deletes the table's structure along with all the data it contains.

Q99 **COMMAND PURPOSE | SELECT**

Which of the following is TRUE about the SELECT statement?

A. It can only be used with GROUP BY

B. It retrieves data without changing it

C. It always deletes rows after retrieval

D. It modifies data in a table

Ans: B

Explanation: SELECT is a read-only DML command that fetches data without altering the table's contents.

Q100 **QUERY-BASED | AGGREGATE**

Which function, combined with WHERE, would best answer "How many students scored above 90 marks?"

A. MAX()

B. SUM()

C. COUNT()

D. AVG()

Ans: C

Explanation: COUNT() combined with a WHERE condition counts the number of rows satisfying that condition.

Q101 **COMMAND PURPOSE | DDL**

Which of the following statements about ALTER TABLE is correct?

A. It only renames a database

B. It permanently deletes a table

C. It creates a brand-new table

D. It is used to add, modify, or drop columns in an existing table

Ans: D

Explanation: ALTER TABLE changes the structure of an existing table without deleting the table itself.

Q102 QUERY-BASED | MIXED

Query: "SELECT Dept, MAX(Salary) FROM Employee GROUP BY Dept ORDER BY MAX(Salary) DESC;" — what does this show?

A. Each department's highest salary, sorted from highest to lowest

B. Each employee's salary sorted alphabetically

C. Departments listed randomly with average salary

D. Only the single highest salary in the entire company

Ans: A

Explanation: GROUP BY finds the highest salary per department, and ORDER BY DESC arranges these results from highest to lowest.

Q103 TRICKY | CLAUSE ORDER

Which is the correct logical order for grouping data, filtering groups, and sorting the final output in a query?

A. HAVING → GROUP BY → ORDER BY

B. GROUP BY → HAVING → ORDER BY

C. ORDER BY → GROUP BY → HAVING

D. ORDER BY → HAVING → GROUP BY

Ans: B

Explanation: Data is first grouped, then groups are filtered with HAVING, and finally the result is sorted with ORDER BY.

Q104 SCENARIO-BASED | UPDATE

Which command should be used to correct a wrongly entered Marks value for one specific student, without affecting other records?

A. DELETE the record then INSERT the whole table again

B. DROP and CREATE the table again

C. UPDATE with a WHERE clause for that student

D. ALTER TABLE

Ans: C

Explanation: UPDATE with a WHERE clause modifies only the targeted record, leaving all other rows unchanged.

Q105 **ASSERTION-REASON | AGGREGATE**

Assertion (A): COUNT(*), SUM(), AVG(), MIN(), MAX() are all classified as aggregate functions in SQL.
Reason (R): Aggregate functions perform a calculation on a set of values and return a single summary value.

A. A is false, R is true

B. A is true, R is false

C. Both A and R are true, but R does not explain A

D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: All five functions summarize multiple values into a single result, which is the defining property of aggregate functions.

Q106 **SCENARIO-BASED | MIXED**

After executing "INSERT INTO Student (RollNo, Name, Class, Marks) VALUES (6,'Priya',9,95);" on the Student table, what will "SELECT COUNT(*) FROM Student;" return?

A. 5

B. 7

C. 6

D. 4

Ans: C

Explanation: Adding one new row through INSERT increases the total row count from 5 to 6.

Q107 **QUERY-BASED | MIXED**

Which query lists only classes with more than 2 students, sorted by class in ascending order, using the Student table?

A. SELECT Class, COUNT(*) FROM Student WHERE COUNT(*) > 2 GROUP BY Class ORDER BY Class;

B. SELECT Class, COUNT(*) FROM Student GROUP BY Class ORDER BY Class HAVING COUNT(*) > 2;

C. SELECT Class, COUNT(*) FROM Student HAVING COUNT(*) > 2 GROUP BY Class ORDER BY Class;

D. SELECT Class, COUNT(*) FROM Student GROUP BY Class HAVING COUNT(*) > 2 ORDER BY Class;

Ans: D

Explanation: The correct order is GROUP BY first, then HAVING to filter groups, and finally ORDER BY to sort the output.

Q108 **QUERY-BASED | MIXED**

Which query lists names of Class 10 students, sorted by Marks in descending order, using the Student table?

A. SELECT Name FROM Student GROUP BY Class=10 ORDER BY Marks DESC;

B. SELECT Name FROM Student HAVING Class=10 ORDER BY Marks DESC;

C. SELECT Name FROM Student ORDER BY Marks DESC WHERE Class=10;

D. SELECT Name FROM Student WHERE Class=10 ORDER BY Marks DESC;

Ans: D

Explanation: WHERE must come before ORDER BY in a SELECT statement; it filters rows first, then ORDER BY sorts the filtered result.